

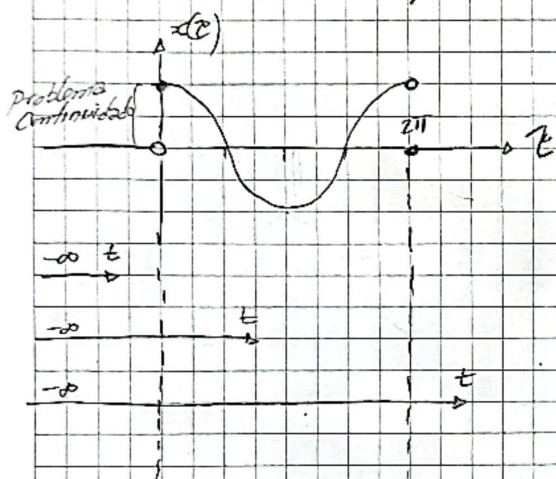
Ejercicio

$$x(t) = \begin{cases} \cos t & 0 \leq t \leq 2\pi \\ 0, & \text{otro caso} \end{cases} \rightarrow \boxed{\int} \rightarrow y(t) = \int_{-\infty}^t x(\tau) d\tau$$

① cambio de variable

$$x(\tau) = \begin{cases} \cos \tau & 0 \leq \tau \leq 2\pi \\ 0 & \text{otro caso} \end{cases}$$

Análisis de la señal y dominio de integración



$$y(t) = \begin{cases} \int_{-\infty}^t x(\tau) d\tau & t < 0 \quad \textcircled{1} \\ \int_{-\infty}^t x(\tau) d\tau & 0 \leq t \leq 2\pi \quad \textcircled{2} \\ \int_{-\infty}^t x(\tau) d\tau & t > 2\pi \quad \textcircled{3} \end{cases}$$

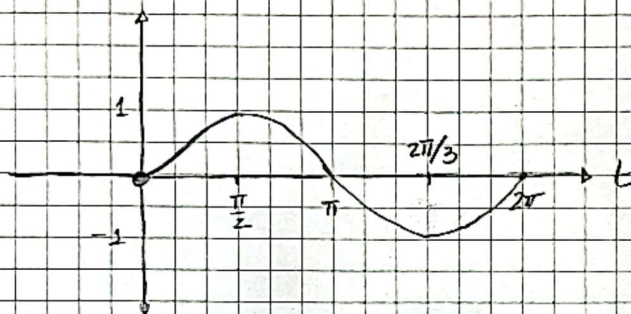
para ① $\int_{-\infty}^t x(\tau) d\tau = \int_0^t 0 d\tau = 0$ para $t < 0$

para ② $\int_{-\infty}^t x(\tau) d\tau = \int_{-\infty}^0 0 d\tau + \int_0^t \cos \tau d\tau$
 $= \sin \Big|_0^t = \sin t - \sin 0 = \sin t$ para $0 \leq t \leq 2\pi$

para ③ $\int_{-\infty}^t x(\tau) d\tau = \int_{-\infty}^0 0 d\tau + \int_0^{2\pi} \cos \tau d\tau + \int_{2\pi}^t 0 d\tau$
 $= \sin \Big|_0^{2\pi} = 0$ para $t > 2\pi$

$$y(t) = \begin{cases} 0 & t < 0 \\ \sin t & 0 \leq t \leq 2\pi \\ 0 & t > 2\pi \end{cases} *$$

* no se conciben las discontinuidades



Ejercicio

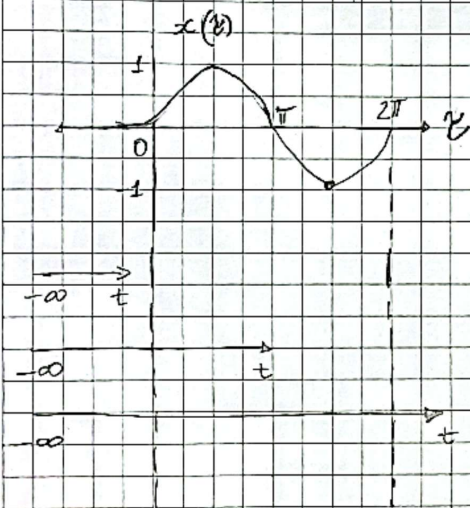
$$x(t) = \begin{cases} \sin(t) & 0 \leq t \leq 2\pi \\ 0 & \text{otro caso} \end{cases} \longrightarrow \boxed{\int} \longrightarrow \psi(t) = \int_{-\infty}^t x(\tau) d\tau$$

① Cambio de variable

$$x(\tau) = \begin{cases} \sin(\tau) & 0 \leq \tau \leq 2\pi \\ 0 & \text{otro caso} \end{cases}$$

$$\psi(t) = \int_{-\infty}^t x(\tau) d\tau$$

② Análisis de la señal y dominio de integración



$$\psi(t) = \begin{cases} \int_{-\infty}^t x(\tau) d\tau & t < 0 \quad \text{①} \\ \int_{-\infty}^t x(\tau) d\tau & 0 \leq t \leq 2\pi \quad \text{②} \\ \int_{-\infty}^t x(\tau) d\tau & t > 2\pi \quad \text{③} \end{cases}$$

para ① $\int_{-\infty}^t x(\tau) d\tau \quad t < 0$

$$\int_{-\infty}^t 0 d\tau = 0 \quad \text{para } t < 0$$

para ② $\int_{-\infty}^t x(\tau) d\tau \quad 0 \leq t \leq 2\pi$

$$\int_{-\infty}^0 0 d\tau + \int_0^t \sin \tau = -\cos \tau \Big|_0^t = -\cos t + 1 \quad \text{para } 0 \leq t \leq 2\pi$$

1 punto

para ③ $\int_{-\infty}^t x(\tau) d\tau \quad t > 2\pi$

$$= \int_{-\infty}^0 0 d\tau + \int_0^{\pi} \sin(\tau) d\tau + \int_{2\pi}^t 0 d\tau$$

$$= -\cos \tau \Big|_0^{\pi} = -(\cos(\pi) + \cos(0))$$

$$= 0$$

para $t > 2\pi$

entonces

$$y(t) = \begin{cases} 0 & t < 0 \\ 1 - \cos t & 0 \leq t \leq 2\pi \\ 0 & t > 2\pi \end{cases}$$

